

CRF Fluctuation Law v3

$$Q_{\text{true}} = \sigma_N / \sqrt{N} = 1 / \sqrt{\kappa n T_{\text{avg}}}$$

Unit correction, 5-seed reconfirmation, and honest scope. Supersedes v1 and v2.

Ougit Jarittum

ORCID: 0009-0009-4451-6811 DOI: 10.5281/zenodo.18977059 CC-BY-NC 4.0

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Abstract

We correct a unit error in CRF FluctuationLaw v1/v2 and confirm the corrected law across 5 independent seeds.

Unit error (v1/v2). The original quantity Q_A used cumR/N (total events per node, units: N_{eff}) as the denominator for mean mass, instead of $\kappa \cdot \text{cumR}/N$ (units: Ω). This introduced a hidden $\sqrt{\kappa}$ factor: $Q_A = \sqrt{\kappa} Q_{\text{true}}$. The prediction $1/\sqrt{n T_{\text{avg}}}$ also lacked $\sqrt{\kappa}$, so the 1.0% gap in v1 was a coincidental cancellation of two equal errors, not a genuine match.

Corrected law. The physically correct quantity is:

$$Q_{\text{true}} \equiv \frac{\sigma_N}{\sqrt{N}} = \frac{\text{std}(N_{\text{eff}})}{\sqrt{\text{mean}(N_{\text{eff}})}} \approx \frac{1}{\sqrt{\kappa n T_{\text{avg}}}}.$$

Reconfirmation (5 seeds, 800 sweeps). $Q_{\text{true}} = 0.16129 \pm 0.00227$ (CV = 1.4%). Prediction: $1/\sqrt{\kappa n T_{\text{avg}}} = 0.15338$. Gap: 5.2%.

The law is confirmed as a stable, seed-independent invariant. The gap of 5.2% (vs. a coincidental 1.0% in v1) reflects the honest accuracy of the derivation — Step 2 (independence of inter-event periods) remains a **Hypothesis**.

Contents

1	The Unit Error	2
1.1	What v1/v2 computed	2
1.2	Numerical verification of the error	2
2	The Corrected Law	2
3	5-Seed Reconfirmation	3
3.1	Protocol	3
3.2	Results	3
4	Status Table	4
5	Open Questions	4

1 The Unit Error

1.1 What v1/v2 computed

In the original analysis, the mean mass denominator was:

$$\bar{\Omega}_A = \frac{\text{cumR}[s]}{N}$$

where $\text{cumR}[s]$ is the total R-events across all nodes up to sweep s . This equals $\text{mean}(N_{\text{eff}})$ — the mean firing count per node — not $\text{mean}(\Omega) = \kappa \cdot \text{mean}(N_{\text{eff}})$.

The quantity actually computed was:

$$Q_A = \frac{\sigma_{\Omega}/\kappa}{\sqrt{\bar{\Omega}_A/\kappa}} = \frac{\text{std}(N_{\text{eff}})}{\sqrt{\text{mean}(N_{\text{eff}})/\kappa}} = \sqrt{\kappa} \frac{\text{std}(N_{\text{eff}})}{\sqrt{\text{mean}(N_{\text{eff}})}} = \sqrt{\kappa} Q_{\text{true}}.$$

Simultaneously, the prediction used $1/\sqrt{nT_{\text{avg}}}$ instead of $1/\sqrt{\kappa nT_{\text{avg}}}$. Both the measured quantity and the prediction were off by $\sqrt{\kappa} = \sqrt{0.15} = 0.387$, producing a spurious 1.0% match.

1.2 Numerical verification of the error

Quantity	v1 value	True value	Ratio
Measured Q (5-seed mean)	0.05880	0.16129	2.74
Prediction	0.05941	0.15338	2.58
Expected ratio ($1/\sqrt{\kappa}$)	—	2.582	—

The measured ratio 2.74 is close to $1/\sqrt{\kappa} = 2.582$ (gap 6%), consistent with the 5.2% gap in the corrected law plus run-length effects (800 vs. 1500 sweeps).

2 The Corrected Law

Fluctuation Law v3 (correct units)

$$Q_{\text{true}} \equiv \frac{\text{std}(N_{\text{eff},i})}{\sqrt{\text{mean}(N_{\text{eff},i})}} \approx \frac{1}{\sqrt{\kappa n T_{\text{avg}}}},$$

where $N_{\text{eff},i}$ is the total number of R-events of node i , $\kappa = 0.15$ (mass coupling), $n = 8$ (SO(3), #16), and $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0] = 35.42$ sweeps (#21a).

Numerical prediction

$$Q_{\text{true}} = \frac{1}{\sqrt{0.15 \times 8 \times 35.42}} = \frac{1}{\sqrt{42.50}} = 0.15338.$$

Physical meaning. Q_{true} is the index of dispersion of the firing-count distribution: how much the number of R-events varies across nodes, normalised to Poisson baseline ($Q = 1$ for independent Poisson). $Q_{\text{true}} \approx 0.16 \ll 1$ means nodes fire in a variance-suppressed, correlated regime.

The suppression factor is $1/Q^2 \approx \kappa n T_{\text{avg}} \approx 42$. Each of $n = 8$ $\text{SO}(3)$ modes must accumulate $T_{\text{avg}} = 35.4$ sweeps of KL drift before firing; mass coupling κ scales the energy per event. These three factors together suppress independent-Poisson variance by $\approx 42\times$.

3 5-Seed Reconfirmation

3.1 Protocol

CRF Hybrid V20.5, $N = 500$, $\kappa = 0.15$, 800 sweeps. Q_{true} computed from actual N_{eff} array at each snapshot (sweeps 200–800, every 10 sweeps). Analysis window: last 40% of snapshots.

3.2 Results

Seed	Q_{true} (mean)	Q_{true} (std)	Late slope
42	0.15938	0.00479	-2.1×10^{-4}
123	0.15965	0.00343	-1.7×10^{-4}
777	0.16089	0.00478	-0.2×10^{-4}
2024	0.16565	0.00416	-2.9×10^{-4}
99	0.16089	0.00369	$+0.1 \times 10^{-4}$
Mean	0.16129	0.00227	stable

Quantity	Value	Source
Q_{true} (5-seed mean)	0.16129 ± 0.00227	simulation
Prediction $1/\sqrt{\kappa n T}$	0.15338	δ -chain
Gap	5.2%	
CV across seeds	1.4%	

The law is a stable invariant: CV = 1.4% across seeds, slopes all $< 3 \times 10^{-4}$. The 5.2% gap from prediction is the honest accuracy of a *Hypothesis*-level derivation (Step 2 unproved).

4 Status Table

Result	Status	Note
$Q_{\text{true}} = 0.16129 \pm 0.00227$ (5 seeds)	Confirmed	CV = 1.4%
Q_{true} stationary	Confirmed	all slopes $< 3 \times 10^{-4}$
Unit error in $v1/v2$	Corrected	hidden $\sqrt{\kappa}$ factor
$v1$ 1.0% gap was spurious	Confirmed	error cancelled error
$Q_{\text{true}} = 1/\sqrt{\kappa n T}$ (5.2% gap)	Hypothesis	Step 2 unproved
Suppression $1/Q^2 \approx \kappa n T \approx 42$	Confirmed	numerical
Q_{true} κ -independent (fixed T)	Falsified	$Q \propto \kappa^{0.274}$
Step 2 independence of inter-event periods	Open	Priority 1
Derive κ exponent 0.274	Open	Priority 2

5 Open Questions

Priority 1: Prove Step 2

Step 2 assumes that inter-event variance contributions are independent across $N = 500$ coupled nodes. Cross-correlations via the $k = 20$ KL-neighbour graph contribute at unknown order. Target: show cross-correlations are $O(\varepsilon_0^2)$, making $Q_{\text{true}} = 1/\sqrt{\kappa n T_{\text{avg}}}$ exact at leading order. The 5.2% gap is consistent with $O(\varepsilon_0)$ corrections.

Priority 2: Derive κ -exponent 0.274

$Q_{\text{true}} \approx 0.103 \times \kappa^{0.274}$ ($R^2 = 0.95$) is an empirical scaling not yet derived from the δ -chain. Single-node drift-diffusion predicts the wrong sign ($\propto \kappa^{-1/6}$), pointing to a collective network mechanism. The stationary invariant $A = s \cdot \text{Var}(\lambda_i)/\langle \lambda_i \rangle$ (slope = 0, $R^2 = 0.00$) and $Q_{\text{true}} = \sqrt{A}$ (exact by algebra) together suggest the path runs through the firing-rate heterogeneity and its dependence on $\theta_{\text{eff}} = K^* \sqrt{1 + \kappa N_{\text{eff}}}$.

The error was found. The law survived it.
 $Q_A = \sqrt{\kappa} Q_{\text{true}}$: two equal mistakes, one apparent precision.
The corrected gap is 5.2%, not 1.0%.
That is a more honest number — and a more useful one.
The δ -chain has not finished speaking.